

ESP32-S3 Voice Assistant Architecture

Overview

This project is an experimental embedded AI voice-assistant platform built around the ESP32-S3 N16R8.

The objective is not simply to flash prebuilt firmware, but to fully understand and control:

- audio capture
- speech detection
- OLED interface logic
- networking
- transcription pipeline
- AI interaction flow

The system is evolving toward a modular architecture where:

```
ESP32-S3
↓
voice detection + recording
↓
WiFi transmission
↓
server-side AI processing
↓
response back to device
```

Hardware

Main Board

- ESP32-S3 N16R8
- 16MB Flash
- 8MB PSRAM

Peripherals

OLED Display

- SSD1306
- I2C
- Address: 0x3C

Microphone

- INMP441 I2S microphone

Audio

- External amplifier + speaker module
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Current Status

Confirmed Working

- ESP32 boot and stability
 - OLED display
 - I2C scanning
 - WiFi communication
 - Flask communication
 - INMP441 microphone
 - live microphone signal analysis
 - speech threshold calibration
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Major Lessons Learned

Prebuilt Firmware Limitations

Many ESP32 chatbot systems:

- use hidden GPIO mappings
- rely on undocumented board profiles
- fail silently when hardware differs
- obscure the actual signal path

This project moved away from blind firmware flashing toward a controlled architecture.

Audio Pipeline Findings

Initial Whisper attempts failed because:

- audio was fragmented
- streaming chunks were too small
- speech boundaries were unclear
- silence detection was unstable

The project now focuses on:

speech detection
→ full utterance recording
→ clean upload
→ transcription

instead of continuous fragmented streaming.

Speech Threshold Testing

Observed microphone ranges:

Silence / Room Noise

100–300

Speech Peaks

7000–20000+

This confirmed:

- strong signal separation
 - valid I2S configuration
 - reliable speech triggering potential
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Current Development Direction

Voice Activity Detection

The ESP32 will:

- wait for speech
- trigger recording automatically
- stop after silence
- transmit complete audio blocks

OLED states:

WAIT
REC
SEND

Planned Features

- stable speech segmentation
- Whisper transcription
- GPT integration
- OLED conversational UI
- audio playback
- local wake-word support
- hybrid local/cloud AI processing

Philosophy

This project treats the ESP32 as:

- an embedded signal-processing platform
- a controllable AI edge device
- a modular experimental architecture

rather than a closed “AI gadget”.

Repository Purpose

This repository documents:

- hardware experiments
- signal-analysis tools
- ESP32 firmware development
- AI integration architecture
- debugging methods
- lessons learned during development

Suggested Repository Names

Technical

- `esp32-s3-voice-core`
- `edge-voice-interface`

- `esp32-ai-audio-node`

Experimental / Research Style

- `signal-to-language`
- `open-voice-edge`
- `embedded-cognition-lab`

Minimalist

- `vox-s3`
- `echo-node`
- `s3-listener`

Current Priority

Finalize:

1. stable voice thresholds
2. speech segmentation
3. clean audio uploads
4. reliable Whisper transcription

before expanding into larger assistant logic.